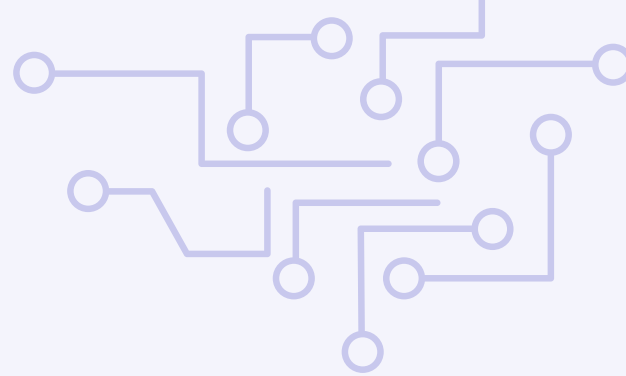
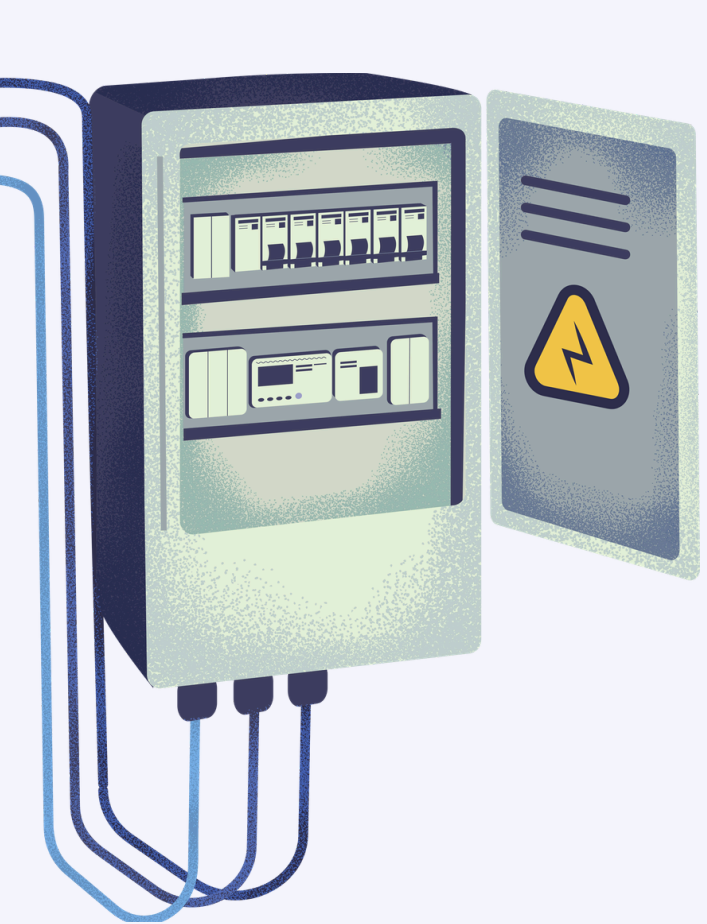
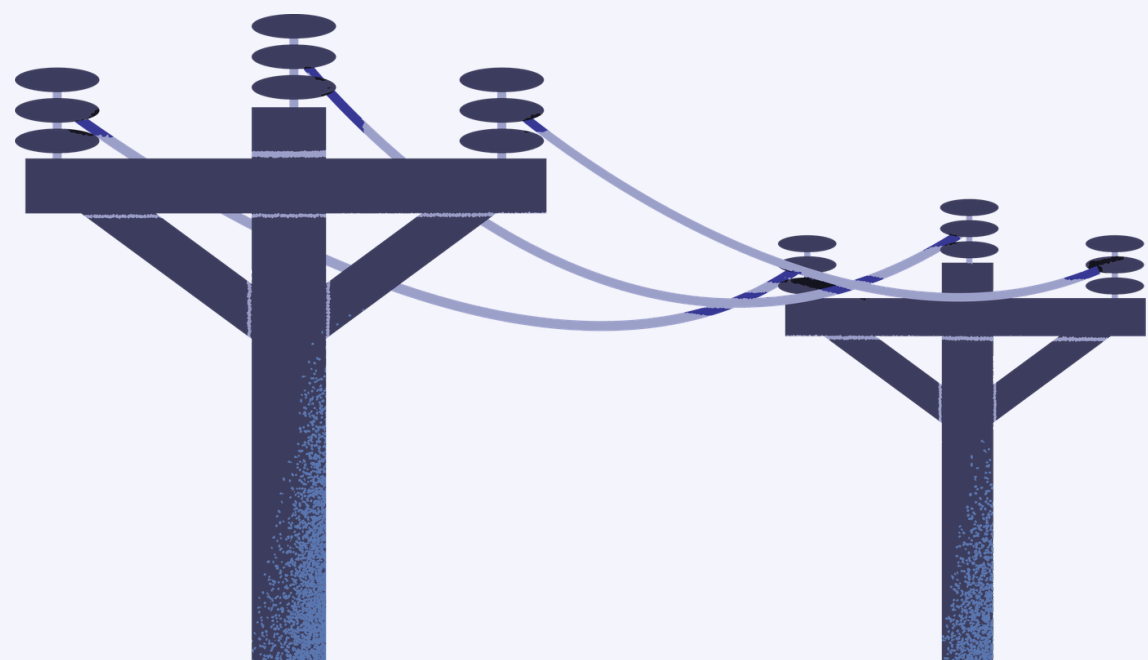


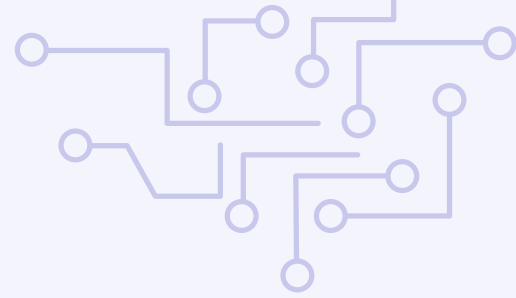
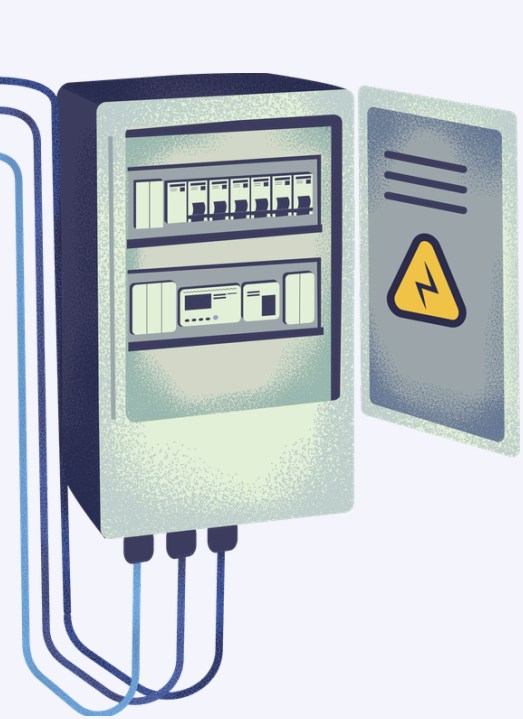


Smart Meter Anomaly Detection

From anomaly detection to power outage reconstruction

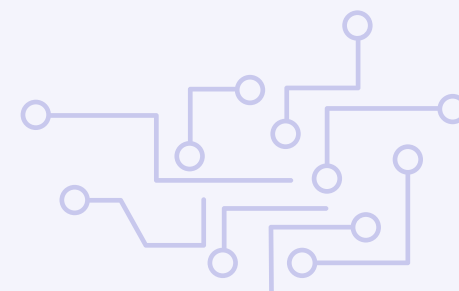
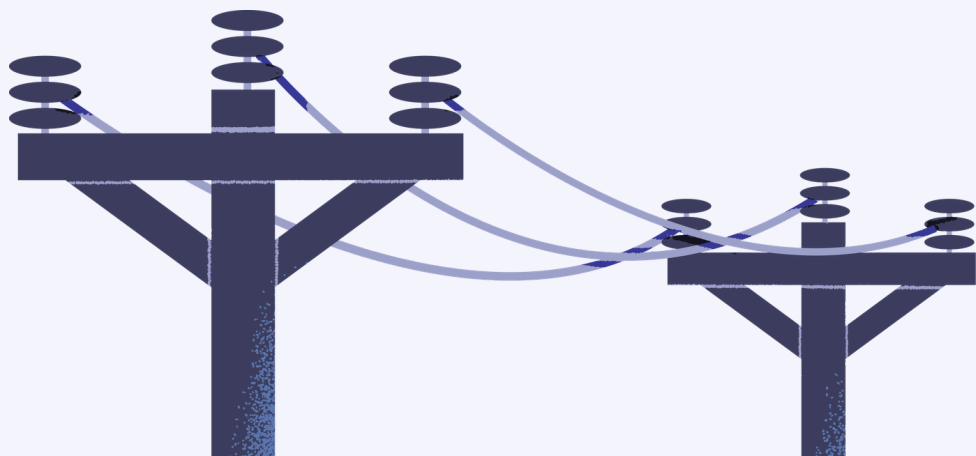
by Nullptr

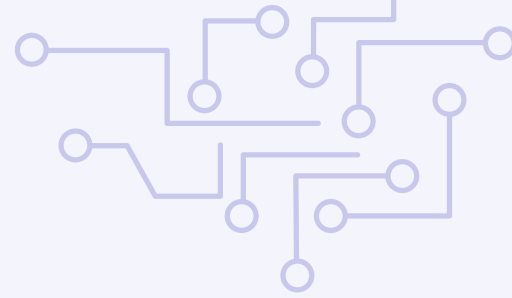
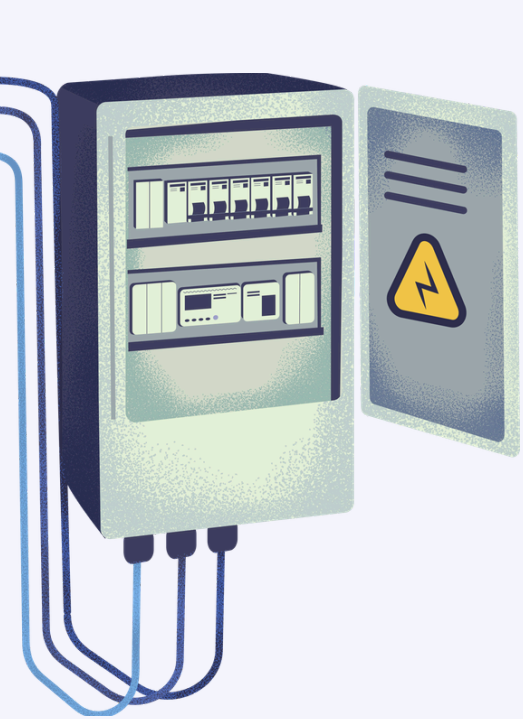




Problem?

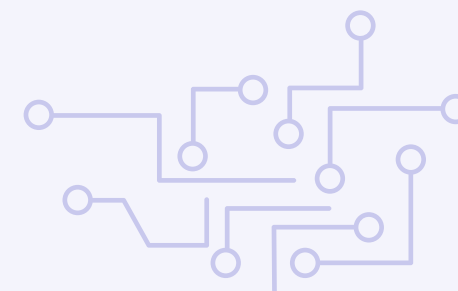
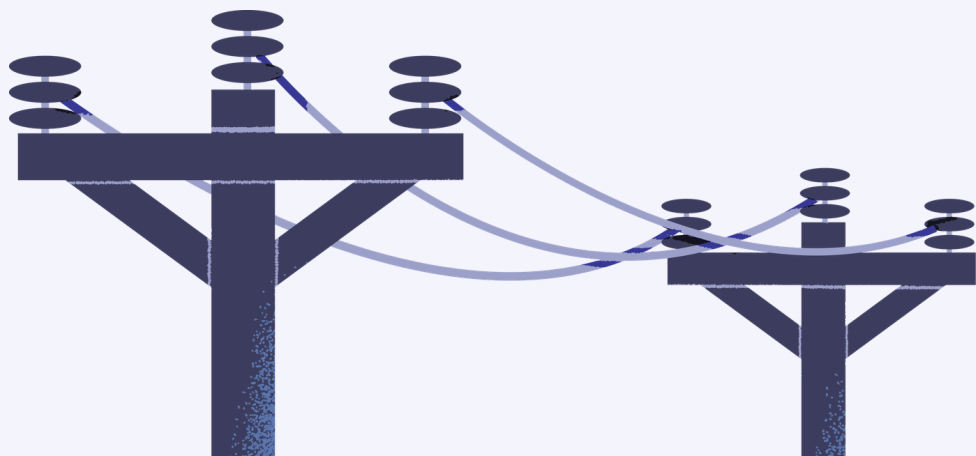
- No real-time visibility into low-voltage grid **events**
- Operators cannot determine **what happened** or **which households** are affected
- We must **reconstruct** events from historical data and **estimate** reliability (SAIDI / SAIFI)





Data?

- **Labeled dataset:** 199 meters, 10 days (~48K rows)
- **Unlabeled dataset:** 200 meters, 7 months (~1M rows)
- **Anomaly rate:** 8.19%
- **Hourly smart meter readings per household**

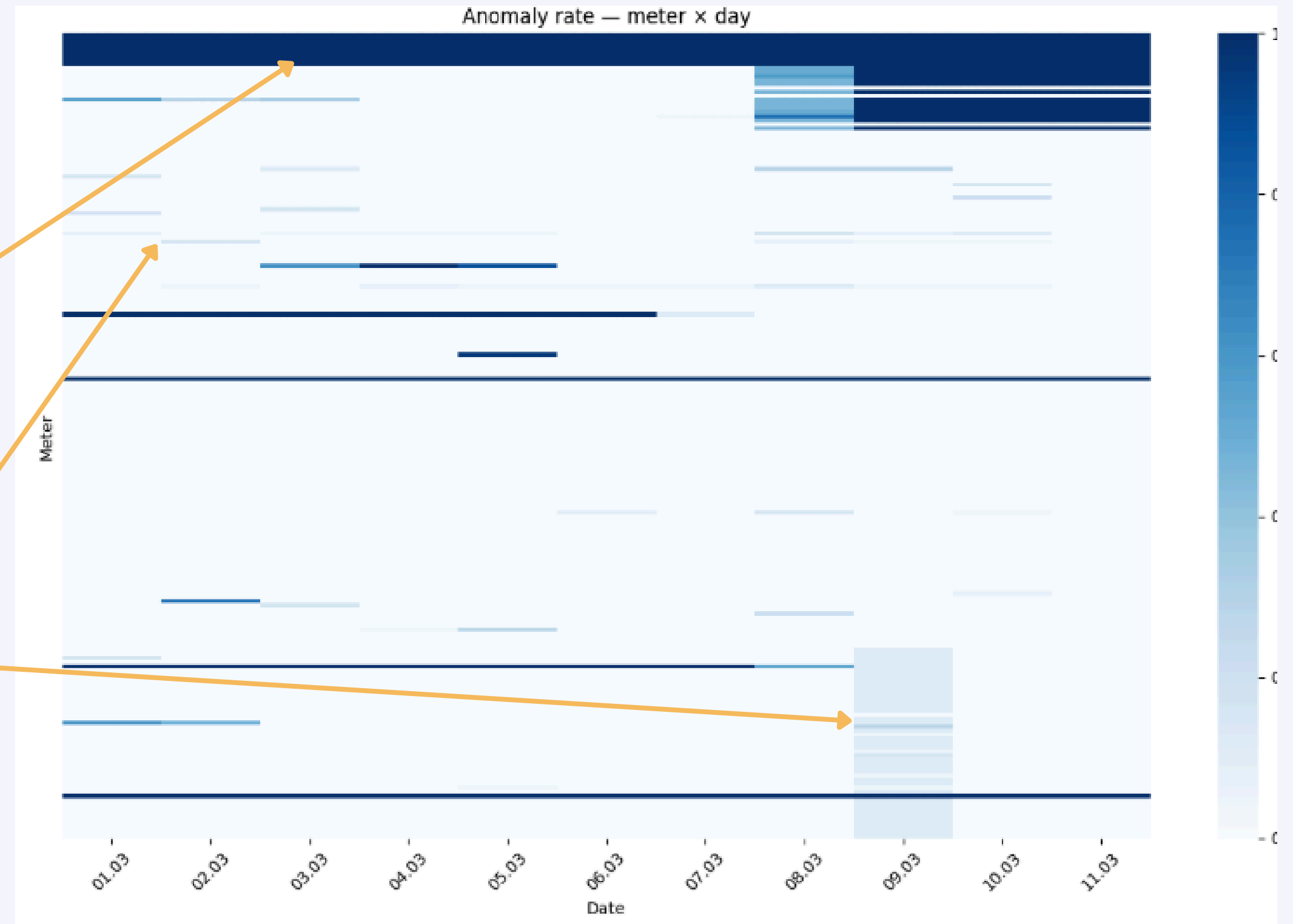


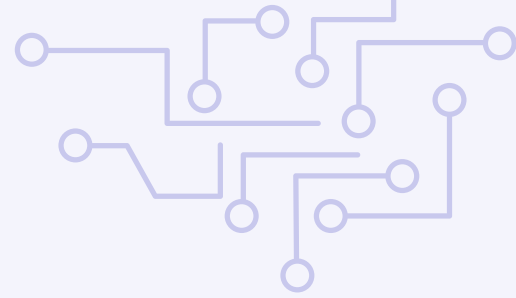
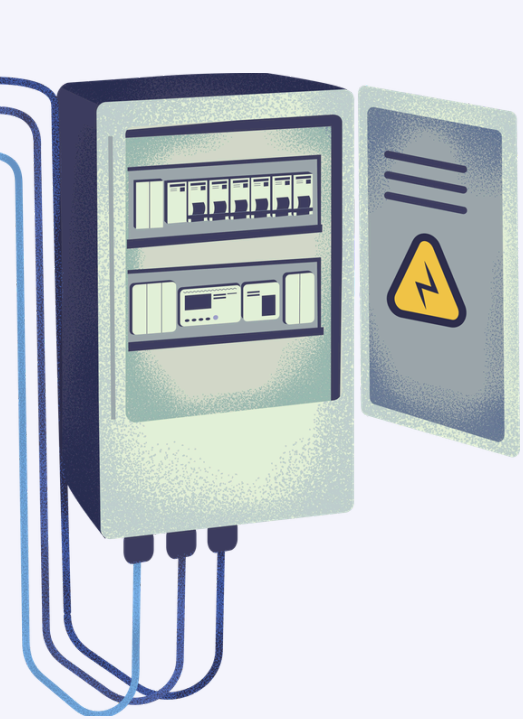
Three distinct event patterns emerge

Broken meters

Outage

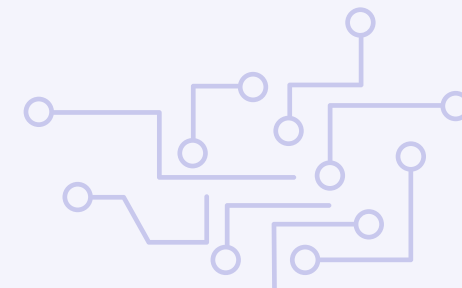
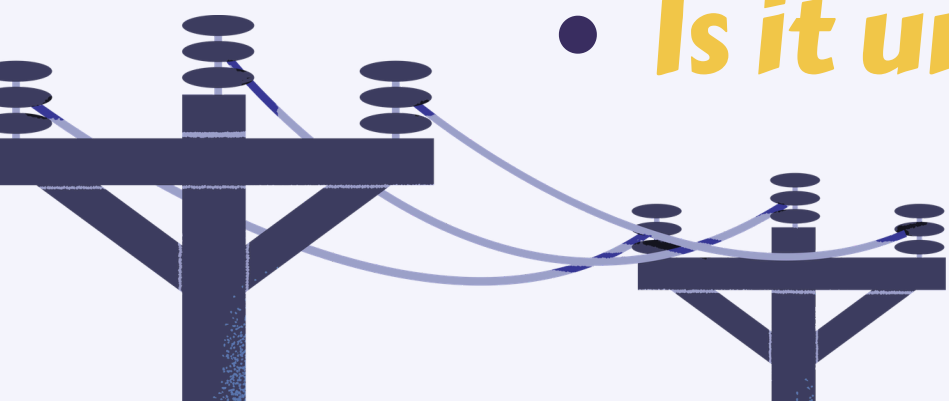
Communication error

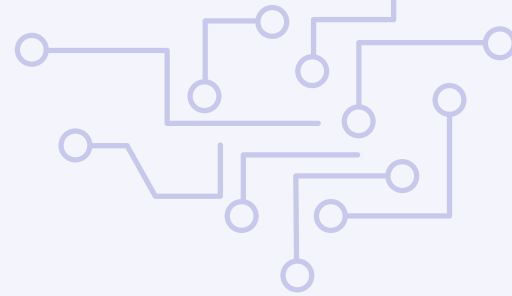
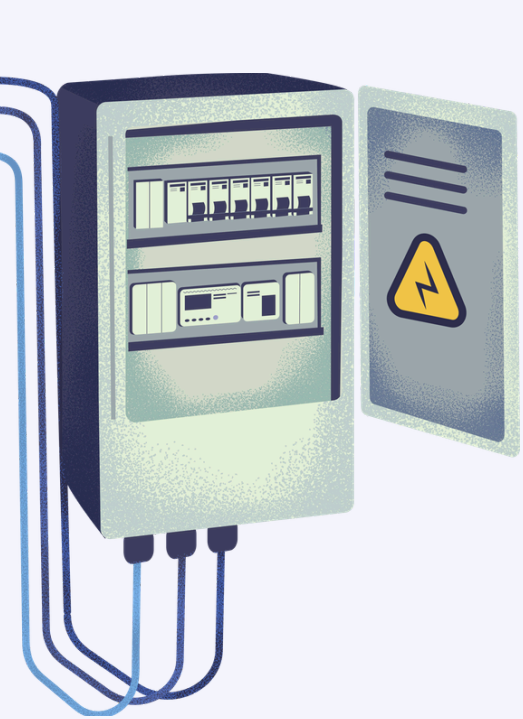




VIP Features?

- **Is it dead?** *is_zero, is_low*
- **How long?** *frozen_streak, meter_zero_rate_72h*
- **Is everyone dead?** *simult_zero_count*
- **Is it stable?** *rolling_std_24h/6h, rolling_mean_24h*
- **Did it change?** *value_diff_1h, sudden_drop*
- **Is it unusual?** *zscore_24h, global_zscore*





So... How Do We Catch the Weird Stuff?

Model

- **LightGBM (final)**
- **XGBoost(baseline)**

Training

- **Meter-level split (80/20)**
- **Prevents data leakage**

Imbalance handling

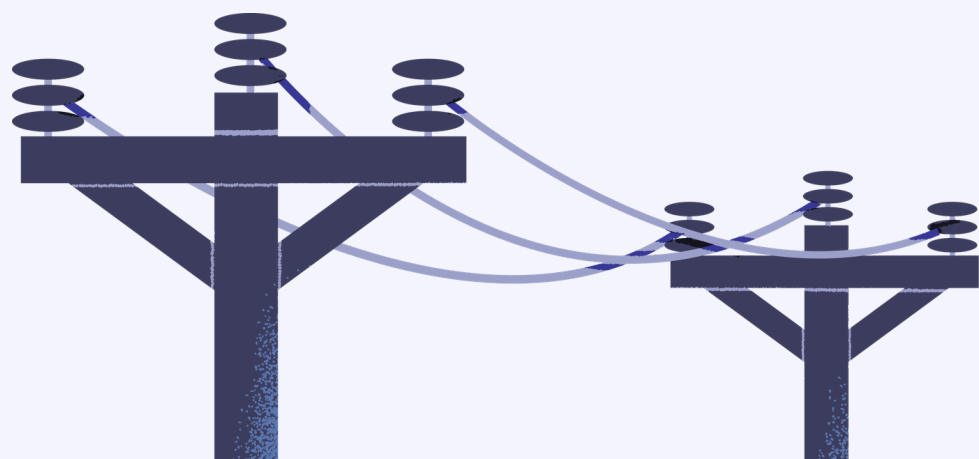
- **scale_pos_weight**
- **Rare anomalies emphasized**
- **Threshold tuned for F1-score**

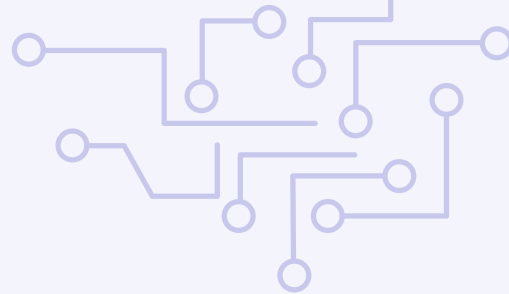
Validation

- **5-fold stratified CV**
- **Consistent performance across meters**

F1 = 0.93 (Precision = 0.95 | Recall = 0.91)

Stratified 5 fold CV F1 : 0.9 ± 0.02





Key insights from 7 months of data

72% communication errors
22% outages
1% broken meters
4% other

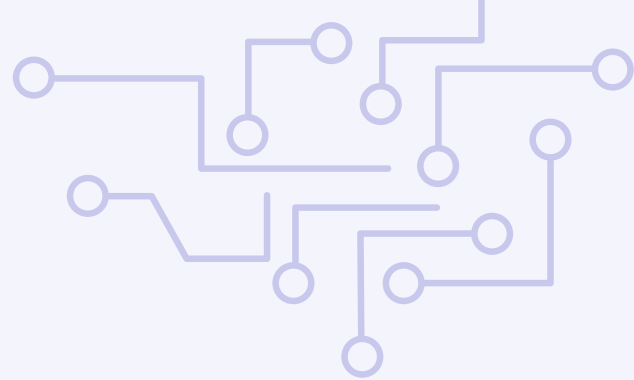
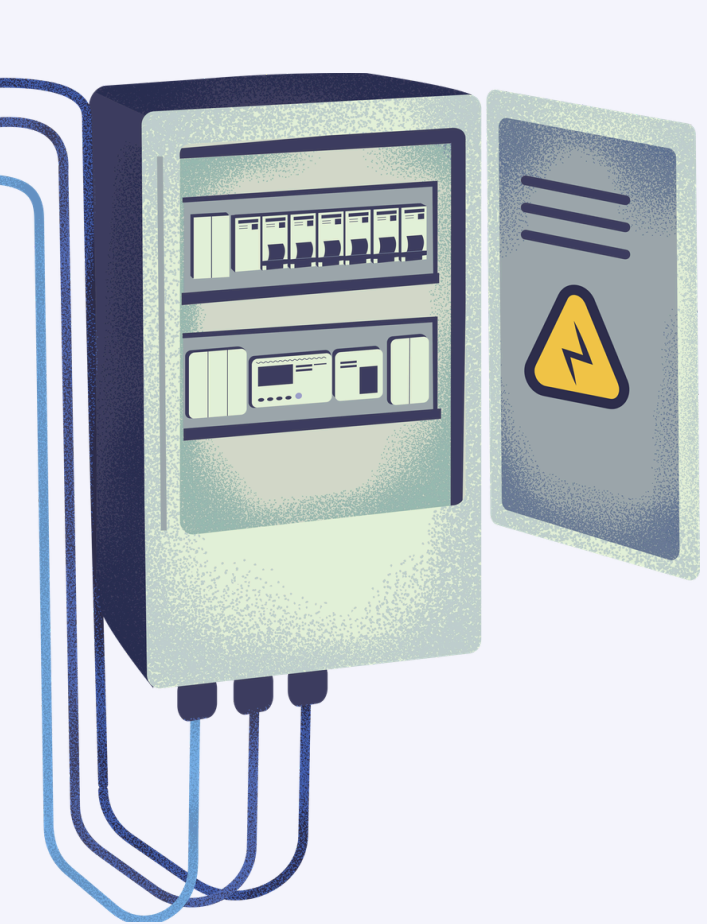
Biggest power outage
2.7. 92 meters 23h 46% grid network

Peak outages: July–August



SAIDI: 384.47 hours per customer
SAIFI: 32.11 interruptions per customer





***Thank
you***

